

XINHAI CHANG

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Education

Peking University, Beijing, China

Expected: Jun 2027

Bachelor of Science in Data Science, Yuanpei College

- GPA: 3.748/4.0
- Coursework: Computer Vision, Machine Learning, Multimodal Learning, Intelligent Robotics [3.9+]

Research Interests

My research bridges **Computer Vision** and **Robotics** to close the loop of **Real-Sim-Real** for Embodied AI. I focus on vision-centric **High-Fidelity 3D Reconstruction and Generation** (Real-to-Sim) to digitize complex physical worlds into simulation-ready assets, and leverage these assets for robotics-centric **Generalizable Robot Policy Learning** (Sim-to-Real) to deploy robust intelligence in the real world.

Technical Skills

- **Robotics and Simulation:** Isaac Lab, Isaac Sim, Curobo, Robot Manipulation, Motion Planning.
- **Languages and Frameworks:** Python, C/C++, Bash, PyTorch, TensorFlow, Pandas, Qt.
- **Algorithms and Vision:** 3D Reconstruction (Gaussian Splatting), Multimodal Learning, 3D Generation.

Research Experience

CoRe Lab, Peking University

Mar 2025 - Present

Research Assistant (Advised by Prof. Yixin Zhu)

- Focusing on **Sim-to-Real** intelligence, specifically large-scale simulation and robust mobile manipulation.

Harvard AI and Robotics Lab

Apr 2024 - Present

Research Assistant (Mentored by Kaichen Zhou)

- Focusing on **Real-to-Sim** pipelines, working on high-fidelity 3D reconstruction and digital twin creation.

Publications

- Niu, Y.*, **Chang, X.***, Liu, X.*, Jiao, Z., Zhu, Y. (2025). *A Scalable Whole-body Trajectory Generator for Coordinated Mobile Manipulation*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
- Li, H.*, Huang, X.*, **Chang, X.***, Zhou, J., Zhao, H. (2025). *TerraX: Visual Terrain Classification Enhanced by Vision-Language Models*. International Conference on Intelligent Robots and Systems (IROS) (Oral).
- Zhou, K.*, Wang, Y.*, Chen, G.*, **Chang, X.**, Beaudouin, G., Zhan, F., Liang, P. P., Wang, M. (2025). *PAGE-4D: Disentangled Pose and Geometry Estimation for 4D Perception*. International Conference on Learning Representations (ICLR).
- Tang, C.*, **Chang, X.*** (2025). *StyleCraft: High-Quality Arbitrary Style Transfer via Unified Content-Style Fusion*. International Conference on Intelligent Computing (ICIC).

Preprints

- Zhou, K.*, **Chang, X.***, Kim, T.*, Zhang, J.*, Cao, Y., Peng, C., Zhan, F., Zhao, H., Dong, H., Ting, K. M., Zhu, Y. (2025). *RAD: A Realistic Multi-View Benchmark for Pose-Agnostic Anomaly Detection*. Under Review.
- Zhou, K.*, Hong, L.*, **Chang, X.***, Zhong, Y., Xie, E., Dong, H., Li, Z., Yang, Y., Li, Z., Zhang, W. (2025). *SplatMesh: Interactive 3D Segmentation and Editing Using Mesh-Based Gaussian Splatting*. Under Review.
- **Chang, X.**, Zhou, K. (2025). *Neural Surface Reconstruction from Sparse Views Using Epipolar Geometry*. Under Review.

* denotes equal contribution

*I. Real-to-Sim: High-Fidelity 3D Reconstruction and Generation***EpiS: Neural Surface Reconstruction from Sparse Views Using Epipolar Geometry** Preprint

- A neural surface reconstruction model resolving geometric ambiguity via epipolar geometry and depth priors.
- Responsible for epipolar feature extraction, monocular depth alignment, and baseline evaluation.
- Roadmap: Achieves high-fidelity 3D geometry reconstruction from sparse visual observations. ●

SplatMesh: Interactive 3D Segmentation and Editing Preprint

- An interactive 3D segmentation and editing algorithm integrating Gaussian Splatting with QEM downsampling.
- Responsible for QEM downsampling, mesh editing pipeline construction, and model training and evaluation.
- Roadmap: Generates editable, memory-efficient digital twins for downstream simulation tasks. ●

PAGE-4D: Disentangled Pose and Geometry Estimation for 4D Perception ICLR 2026

- A feedforward 4D perception model featuring a dynamics-aware mask to disentangle static and dynamic scenes.
- Responsible for novel view synthesis module, LVSM architecture adaptation, and baseline evaluation.
- Roadmap: Extends high-fidelity reconstruction to dynamic environments for 4D simulation. ●

RAD: Realistic Benchmark for Pose-Agnostic Anomaly Detection Preprint

- A realistic, robot-captured multi-view benchmark for anomaly detection on reflective and symmetric objects.
- Responsible for dataset curation, 2D and 3D model training, and baseline evaluation.
- Roadmap: Benchmarks high-fidelity perception against unstructured physical complexities. ●●

*II. Sim-to-Real: Generalizable Robot Policy Learning***AutoMoMa: Scalable Coordinated Mobile Manipulation Trajectory Generation** CVPR 2026

- A scalable data generation pipeline utilizing GPU-accelerated planning for coordinated mobile manipulation.
- Responsible for scene generation, large-scale simulation data collection, and policy training and evaluation.
- Roadmap: Generates large-scale manipulation data to enable generalizable policy learning. ●

TerraX: Visual Terrain Classification Enhanced by VLMs IROS 2025

- A vision-language framework combining multimodal data and a CLIP-based VLM for terrain classification.
- Responsible for heterogeneous data organization, fine-grained annotation, and baseline evaluation.
- Roadmap: Enables robust visual terrain perception for generalizable robotic deployment. ●●

Honors and Awards

- **2025 National Scholarship** (Highest honor for undergraduate students in China)
- 2024 Second-class Scholarship, Peking University
- First Prize, 21st Jiang Zehan Cup Mathematical Modeling Competition, 2024
- Xing Zhengde Scholarship, Yuanpei College, 2023
- Third-class Scholarship, Peking University, 2023

Teaching Experience**Teaching Assistant**, Introduction to Computation B (Prof. Jun Sun) Autumn 2025

- Led weekly programming labs, graded assignments, and designed exam questions.

Teaching Assistant, Introduction to Computation C (Prof. Baobao Chang) Autumn 2025

- Led weekly programming labs, graded assignments, and designed exam questions and the final project.

Teaching Assistant, Introduction to Computer Systems (Prof. Lu Zhang) Autumn 2024

- Led weekly discussion sessions, graded assignments, and contributed to exam design.

Extracurricular Activities

- Volunteer Teacher, International Asian Liver Center, Gansu, China, Summer 2024
- Volunteer, Yunnan Golden Monkey Conservation, Peking University, Winter 2023